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Investigation of the Characteristics of Gas Jets Produced by Annular Atomizer

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Abstract. The aim of the work was to study jet flows produced by gas-gas ejector with annular primary flow nozzle and central duct for secondary flow in case of its consequent usage as a liquids atomizer. A complex experimental and numerical study of a coaxial jet has been performed. A set of probe methods for diagnosing flows provides new data on the characteristic of the flow: the effective area of the ejector nozzle critical section, the gas velocity profiles and the ejection coefficient. Experimental data on the coaxial flows are extremely important for the verification of developing theoretical models of liquid destruction by a gas flow using annular ejectors.

INTRODUCTION

Different modern engineering technologies rely on liquid atomization in different applications: burning liquid fuels in different power plants, cooling hot gases (used in chemical, fuel and other industries), obtaining vapor-gas mixtures, creating coatings, producing powders and fibers; liquid atomization is one of the fundamental research fields of multi-phase flow physics [1-4]. Annular ejectors are used for air-stream atomization: a free immersed pressurized jet of gas effluxes from the nozzle and directly impacts the liquid flow forming a gas-liquid torch. When the gas-liquid flow is formed, the droplets in the gas flow are destroyed, and the following factors affect this process: liquid viscosity and surface tension, the characteristic scale of the intermixing zone and the relative speeds of gas and liquid. The design of these type atomizers rules out the contact of aggressive atomized media with the device's parts.

Physical understanding of the processes of non-Newtonian fluids dispersion in the air flow is necessary for selecting rational modes of annular atomizer operation. As of yet, this understanding has not been formed. The situation is aggravated by the absence of a unified theory that would generalize the research findings in the investigation of the flow of gas jets generated by annular ejectors [5, 6]. In the recent years, there have been attempts to research these processes using state of the art means of numerical analysis; this is mostly done for optimizing the parameters of atomizers geometry and reaching highest possible entrainment factors [7-9].

Thus, the purpose of this paper is the numeric and experimental research of the main parameters of jet flows produced by annular atomizer used for spraying liquids by exposure to air flow.

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EXPERIMENT AND THEORY

The law of conservation of mass and momentum is valid for the primary (ejecting) and secondary (ejected) flows:

$$G_3 = G_1 + G_2; \quad \left(\int_{F_3} P_3 dF_3 + G_3 v_3 \right) = \left(\int_{F_1} P_1 dF_1 + G_1 v_1 \right) + \left(\int_{F_2} P_2 dF_2 + G_2 v_2 \right), \tag{1}$$

where G is the flow rate, v is the velocity, P is the pressure, F is the effective flow area, indexes 1 and 2 correspond to the primary and secondary flow, respectively, index 3 corresponds to the resulting (mixed) flow.

Ejection coefficient is one of the main parameters of the ejector, characterizing the efficiency of the operating mode of the device. Ejection coefficient is the ratio of expenses primary and secondary flows

$$K_{\text{ej}} = \frac{G_1}{G_2} \text{ or } K_{\text{ej}} = \frac{G_3 - G_2}{G_2} = \frac{G_3}{G_2} - 1. \tag{2}$$

The flow rate of the mixed flow was determined in a result of integration of the velocity diagram in the selected section of the jet:

$$G_3 = \pi\rho \int_{-r_0}^{r_0} u(r) r dr , \tag{3}$$

where $u(r)$ is the radial distribution of the axial velocity, r_0 is the stream radius, ρ is the air density.

Computational fluid dynamics (CFD) simulations was carried out using commercial software SolidWorks Flow Simulation 2016 (SWFS). Verification of the chosen computation technology was carried out according to the experiments of T. Trupel, as well as O.V. Yakovlevsky and V.K. Pechenkin, and original study results. These experiments were a velocity measurement in an axisymmetric immersed turbulent subsonic stream of circular cross-section. O.V. Yakovlevsky and V.K. Pechenkin considered the dependence of the axial velocity of the jet from the distance to the nozzle edge. The velocity profiles in the cross-sections of the jet were measured by T. Trupel. The results of the experiments were taken from [10].

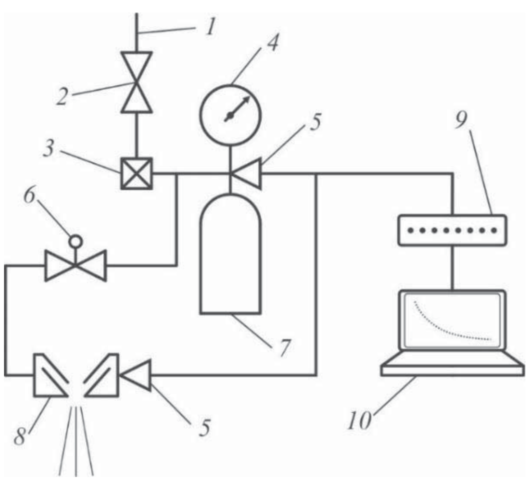


FIGURE 1. Scheme of the experiment for determining the ejector nozzle flow rate coefficient: 1 - compressed gas pipeline; 2 - valve; 3 - reducer; 4 - manometer; 5 - primary measuring transducer of full pressure (PMTFP); 6 - the electromagnetic valve; 7 - cylinder; 8 – ejector [11]; 9 - analog-to-digital converter (ADC); 10 - personal computer (PC)

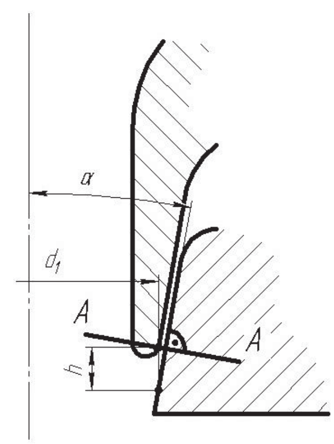


FIGURE 2. Ejector (8) annular nozzle with central duct

The effective area of the ejector nozzle critical section, which is the ratio of the cross-sectional areas of the flow passing through the nozzle and the nozzle section, was carried out experimentally according to scheme (Fig. 1).

Air under high pressure was pumped into a steel cylinder, from which air subsequently was fed into the ejector pre-chamber and freely flowed through the annular nozzle, critical section area of which was controlled by the mutual axial displacement (*h*) of the ejector parts (Fig. 2). The air flow is assumed to be adiabatic and isentropic. The change in the pressure in the cylinder under critical conditions (at $\frac{P_{atm}}{P_0} = \left(1 + \frac{\chi - 1}{2} M^2\right)^{\frac{\chi}{\chi - 1}} \approx 0.528$, $M = 1$) is [12]:

$$P(t) = P_0 \left(1 + Bt\right)^{\frac{-2\chi}{\chi - 1}}, \tag{4}$$

where $P(t)$ is the current cylinder pressure, $B = \frac{\chi - 1}{2V} \mathrm{R}\, mF\mu\sqrt{T_0}$, $m = \sqrt{\frac{\chi}{\mathrm{R}}\left(\frac{2}{\chi + 1}\right)^{\frac{\chi + 1}{\chi - 1}}}$, V is the volume of the cylinder, R is the specific gas constant, F is the critical area, μ is the flow coefficient, T_0, P_0 is the initial temperature and pressure in the cylinder, χ is the adiabatic index, P_{atm} is the atmosphere pressure, M is the Mach number.

The value of B was determined from the results of approximation of the experimental curves of the time dependence of the pressure in the cylinder under the critical flow regime. Thus, the cross-sectional area of the flow passing through the nozzle:

$$\mu F = \frac{2VB}{(\chi - 1)\mathrm{R}\, m\sqrt{T_0}}. \tag{5}$$

The air flow through the nozzle was determined according to the dependence

$$G_0 = m\frac{P_{pr}\mu F}{\sqrt{T_0}}, \tag{6}$$

where P_{pr} is the pre-chamber pressure.

The local axial gas velocity was determined from the measured total and static pressures in the flow sections from the adiabatic equations and the Bernoulli law under the assumptions that heat sources in the gas, as well as external heat sources, are absent, the heat capacities and flow temperatures in the ejector are constant, pressure and temperature satisfy the adiabatic flow of an ideal gas, according to the relation

$$u = \sqrt{\frac{2\mathrm{R}T\chi\left[\left(\frac{P}{P_0}\right)^{\chi - 1/\chi} - 1\right]}{\chi - 1}} \tag{7}$$

where P and P_0 are the total and the static pressure respectively.

During the experiments, the pressure in the ejector pre-chamber was varied in the range P_{pr} = 2-4 atm., and the effective nozzle area - μF = 8.2-22.1 mm^2, the possibility of overlapping the central duct was provided. The sensing element moved in the cross sections of the stream at a speed of v_{sensor} = 5 mm/s, the interrogation of the ADC channels occurred at a frequency of 100 Hz.

RESULTS AND DISCUSSION

The modeling of an axisymmetric subsonic jet effluxing into the immersed space with an average velocity u = 87 m/s through a round nozzle 90 mm in diameter (T. Trupel's experiments) and a similar jet that flows at a velocity u = 90 m/s through a round nozzle 5 mm in diameter (experiments of O.V. Yakovlevsky and V.K. Pechenkin) has been provided. A comparison of the computation results with the experimental data is shown in Figs. 3 and 4, and Table 1.

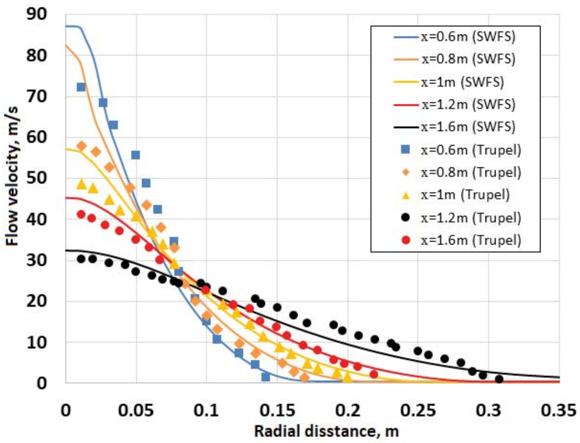


FIGURE 3. Profiles of an axial velocity in cross-sections of the stream: symbols - experiments of T. Trupel; lines - computation results

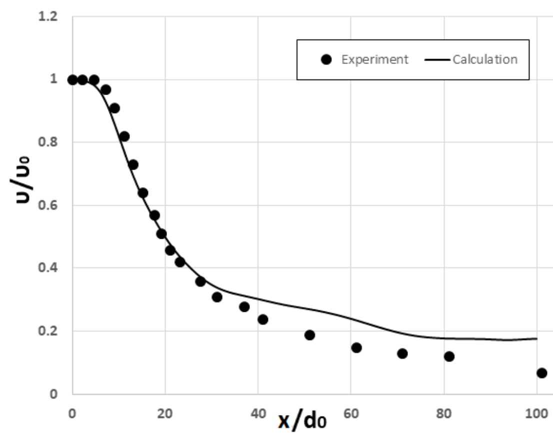
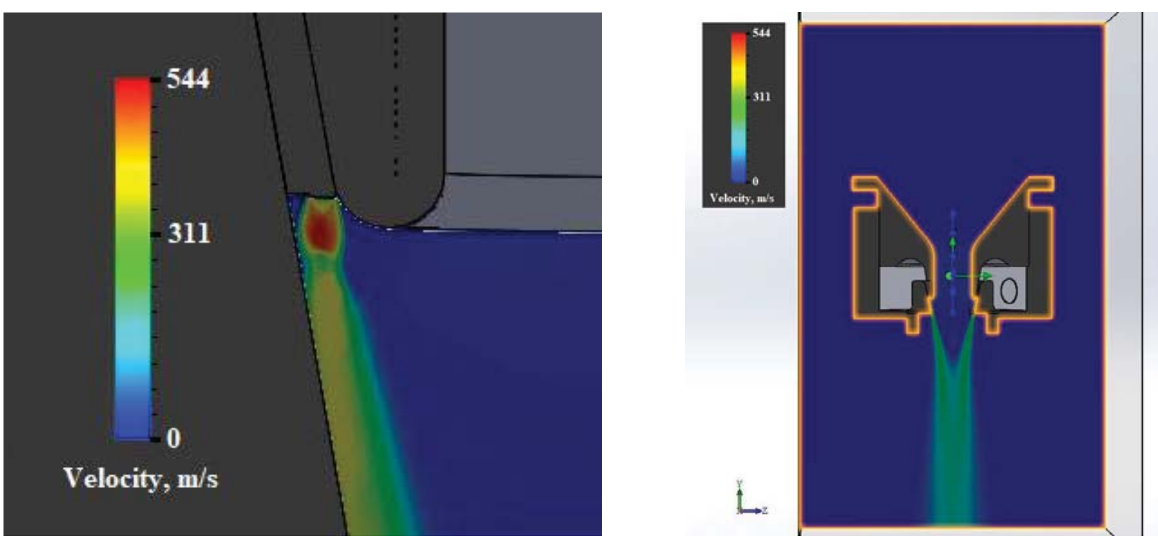


FIGURE 4. Stream velocity along the axis: symbols - experiments of O.V. Yakovlevsky and V.K. Pechenkin; lines - computation results

TABLE 1. Comparison of flow rates based on computation results (SWFS) and the results of [10] for the Trupel problem

X, m	G(Trupel), kg/s	G(SWFS), kg/s	Error, %
0.6	28.8	28.4	1.25
0.8	32.3	34.2	5.61
1.0	39.8	40.4	1.48
1.2	43.7	47.0	7.14
1.6	55.8	61.9	9.88

Represented data shows that the correlation between experiment and calculation is good for all sections under consideration.



(a) (b)

FIGURE 5. Flow patterns obtained by numerical simulation: part of the flow pattern near the ejecting flow nozzle (a); general view of flow pattern (b)

In this simulation of the ejector, the jet flow was assumed to be steady, controlled by the RANS equations together with the continuity equation. The standard k-epsilon turbulence model was adopted to govern the turbulent characteristics with the modified wall function adopted to resolve the near-wall region. The modified wall function uses a Van Driest's profile. In the simulation, flow rate and temperature boundary conditions were applied on the primary flow inlet boundary, and pressure boundary condition on secondary flow inlet and outlet boundaries. The governing equations were discretized by the finite volume method, and the secondary upwind scheme is adopted for spatial discretization of the convection terms. The SIMPLE algorithm was employed to solve the coupling the pressure and velocity. The mesh is three-dimensional, the calculating area covers the entire model of the ejector. The mesh size is initially made a few hundred thousand with thickenings in the area near the flow outlet, and later increased to about 1.5 million to confirm that the results are independent of the mesh. SWFS 2016 software was used in this study.

Figure 5 shows the flow pattern obtained as a result of computations. Verification of the developed model was carried out on the basis of experimental data obtained for the corresponding regimes.

The results of the experiments to determine the effective area of the critical section of the nozzle according (4) and (5) are shown in Fig. 6. In the range of displacement values h from 0.5 to 4.5 mm, the curve (Fig. 6) is well approximated by the polynomial dependence μF=1,05+7,88h+0,65h^2.

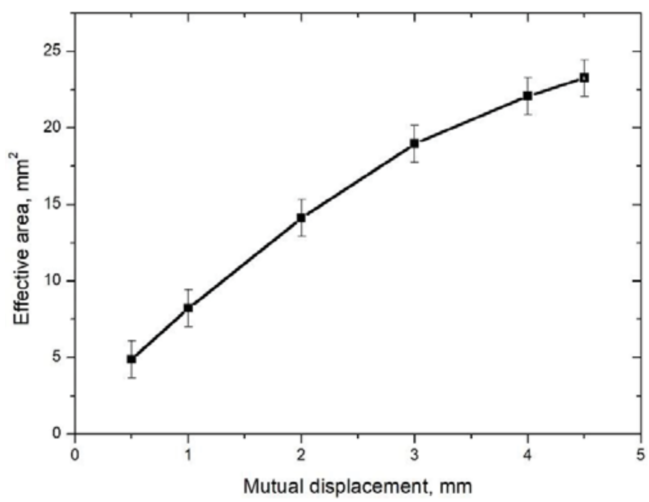
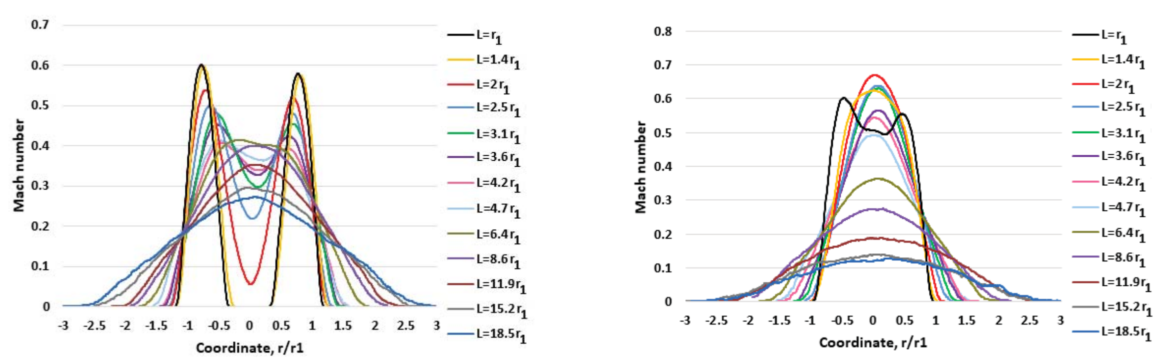


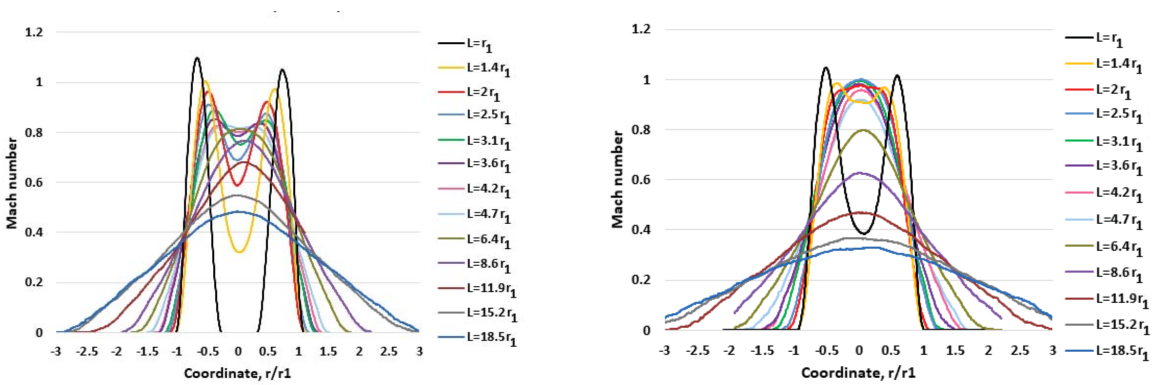
FIGURE 6. The effective area of the ejector nozzle critical section
as a function of the mutual displacement of surfaces forming an ejector nozzle



(a) (b)

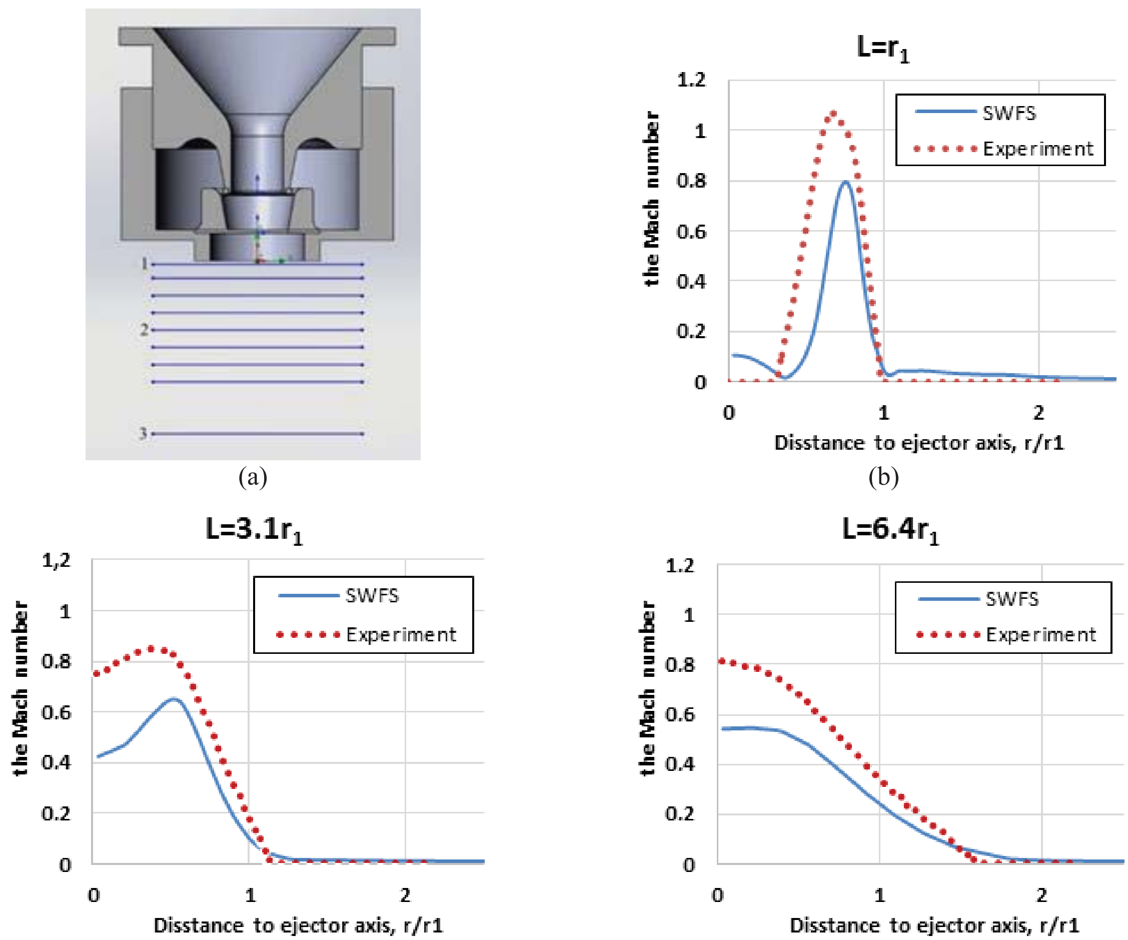
FIGURE 7. The profiles of the average longitudinal velocity in the flow cross-sections at μF = 11.3 mm^2, P_{pr} = 4 atm:
(a) - the central duct is open; (b) - the central duct is closed

The average air longitudinal velocity in cross-sections of flows has been determined in accordance with expression (7). Figures 7 and 8 show the velocity profiles determined for P_{pr} = 4 atm with an open and closed central duct, and for the effective area of the critical nozzle section: μF = 11.3 mm^2 (Fig. 7) and μF = 22.1 mm^2 (Fig. 8).



(a) (b)

FIGURE 8. The profiles of the average longitudinal velocity in the flow cross-sections at μF = 22.1 mm^2, P_{pr} = 4 atm: (a) - the central duct is open; (b) - the central duct is closed



(a) (b) (c) (d)

FIGURE 9. Comparison of computed velocity profiles with experimental: L is the longitudinal distance between the ejector end and the sensor in the experiment; (b) - L=1.0r_1; (c) - L=3.1r_1; (d) - L=6.4r_1; (a) - view of the ejector model with the levels where the computed and experimental velocity profiles were compared.

Comparison of velocity profiles in the cross-sections of the flow, have been computed using the developed mathematical model and have been determined experimentally, is shown in Fig. 9.

Satisfactory agreement between the results of numerical simulation and experimental data on the measurement of the average longitudinal velocity in the flow sections has been reached; the divergence do not exceed 30%.

Analysis of the profiles of the average longitudinal velocity in the cross-sections of the flow provides an evidence the change in the vector field of the flow velocity. With increasing distance to the nozzle edge, the flow velocity profiles being aligned with a tendency to self-similar. So, when the central duct is open, the annular jet is transformed into a continuous one with a round section at a distance to nozzle edge more than $6.4r_1$ (Fig. 7, a). With the central duct closed, occurs an earlier formation of a mixed stream with a larger axial velocity component. Supposedly due to a drawing of the peripheral air.

With an approximately double increase in the effective cross-sectional area of the annular nozzle (Fig. 8), an increase of an air consumption leads to increase the absolute values of the velocity in the flow, but no changes in the character of the flows have been observed. When the central duct is open, the annular jet is transformed into a continuous round one at the same distance to nozzle edge more than $6.4r_1$. With a closed central duct (Fig. 8, b), a slightly later formation of a mixed stream takes place than shown on (Fig. 7, b). The mixed flow is formed at a distance more than $1.4r_1$.

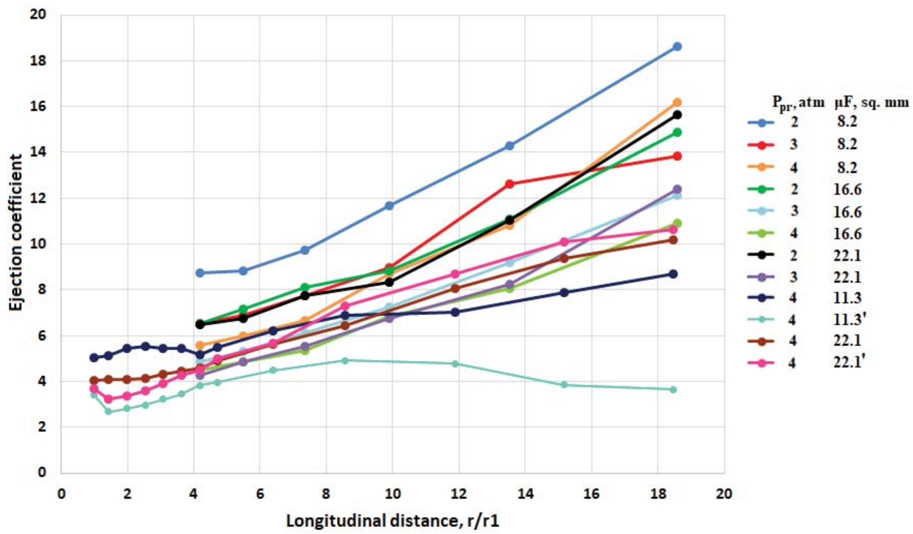


FIGURE 10. The dependence of the ejection coefficient on the pre-chamber pressure P_{pr}, efficient area of the nozzle (μF) and the longitudinal distance: ' mean that central duct closed

The ejection coefficient (Fig. 10) has been determined according to (1-3, 6) over the entire range of variation of the experimental factors (P_{pr} = 2-4 atm., μF = 8.2-22.1 mm^2). It has been established that an increase in even one of factors leads to an increase in the primary flow component in the balance (1); the secondary flow component is increased too. However, the ratio (2) is steadily decreasing, which indicates a more moderate increase in the component of the secondary flow. The mixed flow obeys the law of momentum conservation, which explains the increase in the ejection coefficient within a certain distance from the nozzle. At considerable distances, the ejection coefficient tends to a constant value. In the absence of a mixing chamber, this flow is a free immersed stream; therefore, the absence of a central duct reduces the ejection coefficient in the vicinity of the nozzle.

CONCLUSION

Jet flows produced by gas ejector with annular primary flow nozzle and central duct for the secondary flow have been studied experimentally and numerically. The flow field of gas along with the properties of the atomized liquid is an important condition for its destruction, which ensures the necessary Weber number for the required dispersion.

In the based on RANS CFD simulations is used the standard k-ε turbulence model with modified wall function, which uses a Van Driest's profile. The measured velocity profiles at the device exit as well as results of T. Trupel, O.V. Yakovlevsky and V.K. Pechenkin, have been compared with the CFD predictions. A reasonable agreement between the experimental data and numerical predictions is observed.

It has been found computationally that a recirculation zone at the outlet of the central channel completely disappears when excluding a secondary stream. Investigations of jet stream pattern in transitional regimes are a topic of further work. The dependence of the effective area of the annular nozzle critical section on its geometric parameters has been established. A complex experimental and numerical study of a coaxial jet has been provided the vector field of velocities and the pattern of jet flow produced by the ejector have been established. The ejection coefficient in the flow sections and its dependence on the pressure and the critical area of the nozzle have been determined. The weak influence of these factors on the value of the secondary flow in the central duct of the ejector has been shown.

Obtained experimental data on the coaxial flows are extremely important for the verification of developing theoretical models of liquid destruction by a gas flow using annular ejectors. Thus, the relevance of the study is not limited to technical applications.

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REFERENCES

1. O. S. Nichiporenko, Yu. I. Naida and A. B. Medvedovskiy, *Atomization of metal powders* (Naukova Dumka, Kiev, 1980), 240 p.
2. A. Aliseda, E. J. Hopfinger, J. C. Lasheras, D. M. Kremer, A. Berchielli and E. K. Connolly, International Journal of Multiphase Flow **34 (2)**, 161-175 (2008).
3. I. A. Lysak, T. D. Malinovskaya, G. V. Lysak, A. I. Potekaev, V. V. Kulagina and D. I. Tazin, Russian Physics Journal **59(10)**, 1581-1588 (2017).
4. A. I. Potekaev, I. A. Lysak, T. D. Malinovskaya and G. V. Lysak, Russian Physics Journal **60(11)**, 2047–2049 (2018).
5. V. Yu. Aleksandrov and K. K. Klimovsky, *Optimal atomizers (theory and calculations)* (Machine Engineering, Moscow, 2012), 136 p.
6. S. Elbeland and N. Lawrence, International Journal of Refrigeration **62**, 1-18 (2016).
7. Q. Zaheer and J. Masud, Journal of Visualization **20**, 777-788 (2017).
8. A. Zakharova, E. Vekhter, A. Shklyar and D. Zavyalov, Communications in Computer and Information Science **754**, 215-230 (2017).
9. L. Qiao, Z. Xiao, Q. Zeng, L. Xiao and X. P. Long, Journal of Mechanical Science and Technology **30(2)**, 585-592 (2016).
10. G. N. Abramovich**,** *The Theory of Turbulent Jets* (Fizmatlit, Moscow, 1960), 715 p.
11. I. A. Lysak, G. V. Lysak and T. D. Malinovskaya, RF Patent No. 2614087 (22 March 2017).
12. V. I. Zvegintsev, *Gazodinamicheskie ustanovki kratkovremennogo deystviya. Ch. 1. Ustanovki dlya nauchnykh issledovaniy,* (Parallel, Novosibirsk, 2014), 551 p.